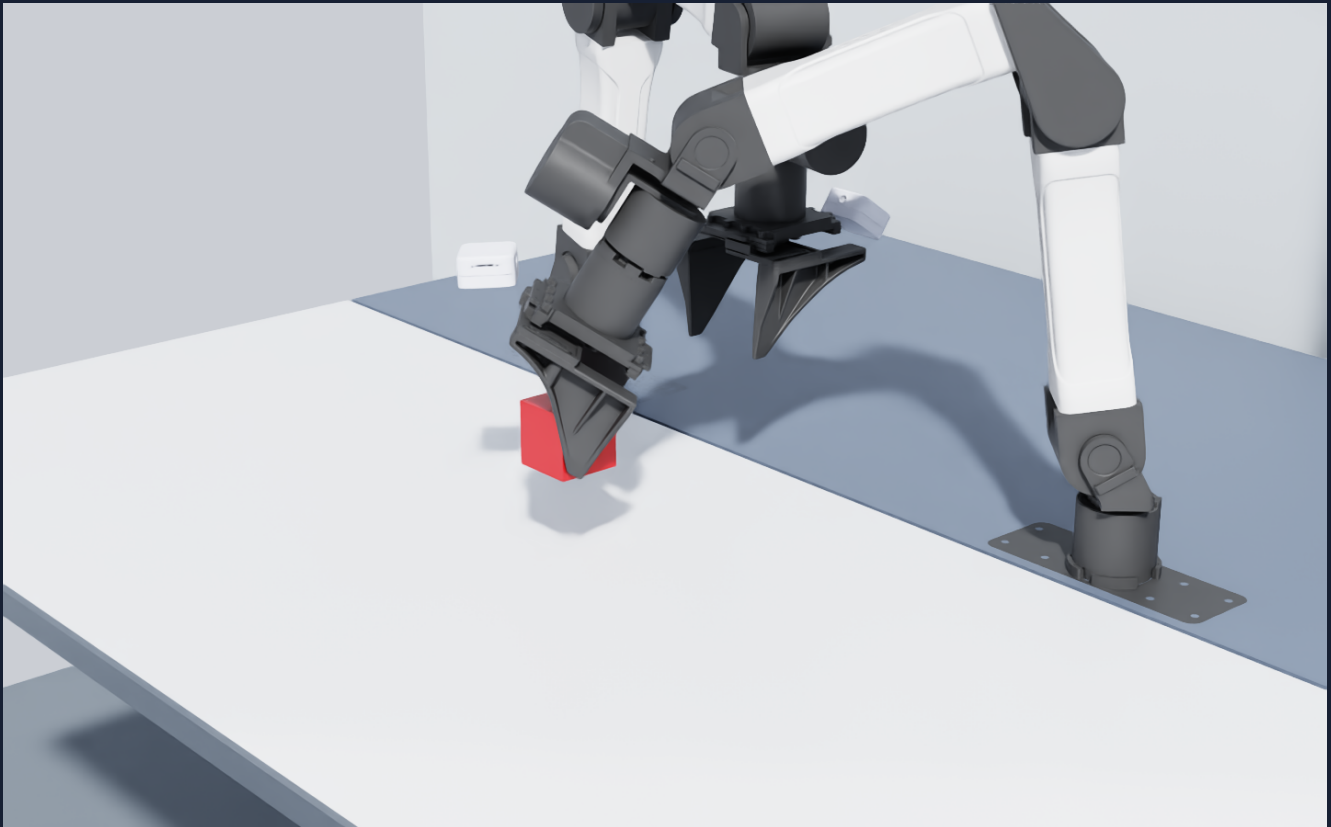


YAM Cube Pick

Physics-First Contact-Only Bimanual Manipulation

Two complete YAM articulations; one dynamic 50 mm cube; finite contact, friction, gravity, and force-limited drives; exact reset-state control; independent verification; numerical convergence, sensitivity, falsification, motion, and qualification evidence

**CALIBRATION STATUS: ENGINEERING / NUMERICAL PRIORS – NOT A
REAL-WORKCELL PREDICTION**



DELIVERABLE

Physical model, equations, source and parameter provenance, installed-engine ownership, robot/control semantics, contact traces, exact reset audit, convergence and sensitivity campaign, retained falsifications, qualification, visual evidence, calibration schema, and reproducibility map.

CURRENT CLAIM

For one hash-bound Isaac Sim 5.1 / PhysX configuration, the left YAM arm physically grasps and lifts the cube from a pre-registered 200 mm square. The cube receives one reset pose write and no later pose writes, joints, attachments, attractors, or hidden forces. The right arm remains simulated but contributes zero commanded action and no cube contact. Material coefficients remain unmeasured priors.

Executive abstract

A credible cube pickup is not established by an animation of a cube following an end effector. The cube must be a freely simulated rigid body; gravity, contact, friction, and robot drive effort must form the complete causal path; the two fingers must make force-bearing contact; lift must persist after table separation; and a verifier must reject right-arm assistance, held or attached objects, post-reset pose authoring, unsafe station contact, excessive speed, and discontinuous control-boundary motion. The scene must also expose what is known, what is merely selected, and what has not been measured.

This dossier records that narrower claim. The source YAM station supplies both six-revolute-joint arms, two-slide grippers, collision shapes, inertias, limits, and force ceilings. Isaac Sim 5.1 and GPU PhysX own the physical state. A 50 mm cube is assigned analytical mass and inertia from a prior density. TGS, patch friction, CCD, a 240 Hz physics step, and a 60 Hz action boundary are authored and read back. A bounded Cartesian controller drives the left articulation through PhysX's measured Jacobian; the right action is exactly zero.

Hash-bound local result. The final evidence set contains 25/25 successful pre-registered qualification episodes, one fixed-rate video, task/overview/contact camera sequences, and a 22-cell matched numerical campaign. Qualification lift spans 80.414 mm–85.126 mm; speed at success spans 0.063738 m/s–0.072452 m/s. Every artifact shares execution-source hash 5ce65c1a57a32f922b5451f369c375cdadd81d31deb294b8902d97668216a718.

Reset-state finding. The first convergence campaign was rejected because equal joint coordinates did not imply equal link state after solver warm-up. USD marker transforms were stale, and fixed-root velocities retained timestep-dependent residuals. The corrected reset reauthors both fixed root poses, sets both root velocities to zero, then reauthors all eight generalized coordinates per arm. Raw PhysX read-back shows all 14 link transforms and all 14 link velocities per arm bit-exact between the matched 120 Hz and 240 Hz probes. Controller and observation poses are now reconstructed from the physical link transform and the authored link-to-grasp transform. The failed campaign is not reused as final evidence.

Boundary-failure finding. A matched seed-588 sweep changed only the pre-close position gate. The 4 mm and 2 mm settings closed too early, made bilateral contact, and lost the cube; the 1 mm and 0.5 mm settings lifted it. The retained 1 mm gate is the least restrictive successful cell and avoids the 21.040 N finger-B peak observed at 0.5 mm. The complete qualification then uses that single setting without seed-specific intervention.

Honest scope. The cube density, table/cube and finger/cube friction, restitution, pad compliance implicit in rigid contact, motor tracking, and the surveyed installation are not measurements from a named physical setup. The current result is a causal, audited simulation result—not a claim that force, slip, trajectory, or success rate match a real YAM cell.

Claim	Status	Authority
Cube motion is produced only by rigid dynamics, contact, gravity, and articulation drives	PASS	one reset write, zero attachments, trace and USD/runtime audit
Both left fingers provide normal-force-bearing contact during successful lift	PASS	substep contact reports and history-sensitive verifier
The right arm assists the pickup	NO	all seven right channels are zero; right-arm/cube contact is a failure
Reset state is matched across numerical cells	PASS	raw PhysX link poses/velocities and one canonical initial-state hash
Canonical result is repeatable and locally converged around the selected step/iterations	PASS	independent repeats and temporal/solver campaign
The model predicts a named physical cube, pad, table, or YAM installation	NO	machine-readable measurement protocol exists; no hardware bundle supplied
Reported contact force includes measured tangential patch-friction impulse	NO	callback exposes normal impulse; tangential patch impulse is unavailable
Two point contacts establish full six-dimensional force closure	NO	only a conservative task-direction support bound is claimed

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Evidence vocabulary. **SOURCE** requested dimension, vendored mechanical asset, official engine behavior, standard, or peer-reviewed model • **LOCAL** reproduced code, read-back, trace, probe, verifier, render, or hash in this repository • **PRIOR** selected engineering value awaiting a named specimen measurement • **REJECTED** implemented or proposed branch that failed its declared causal/numerical gate • **OPEN** missing physical measurement, implementation, convergence, or held-out validation.

A local pass establishes only what happened for an exact execution source and protocol. It cannot promote an unmeasured coefficient into a material property or a simulator trace into hardware data.

1 Fidelity contract: what “pickup” must mean

1.1 Causal task definition

The initial cube center is sampled uniformly in

$$\mathcal{R} = [0.50, 0.70] \times [-0.10, 0.10] \times \{0.775\} \text{ m.}$$

It begins on the 750 mm-high table. A successful history must contain left bilateral contact, table separation, at least 50 mm cube-center lift, and a 0.20 s dwell while the cube remains slow and free of prohibited contacts. The right arm must not touch the cube. The moving left arm must not load the table. The task is intentionally atomic: it tests reach, alignment, finite-aperture closure, contact-supported lift, and stabilization, not transport or placement.

1.2 Anti-cheat invariants

- F1.** The cube is one dynamic rigid body with collision, finite mass, finite inertia, and gravity. It is not parented to, attached to, or constrained to the robot.
- F2.** The cube is authored once at reset. No successful episode may contain a subsequent cube pose write.
- F3.** A policy commands bounded Cartesian and aperture rates. The controller commands force-limited articulation drives, not cube motion or realized link poses.
- F4.** Both fingers must make force-bearing normal contact. A proximity marker, one-sided touch, or table-supported cube cannot satisfy the predicate.
- F5.** Right-arm action is exactly zero in the reference demonstration, and any right-arm/cube contact rejects success.
- F6.** Success is history-sensitive: a prior bilateral grasp and a continuous dwell are required. A single favorable frame cannot pass.
- F7.** Runtime settings, material bindings, solver choice, collision offsets, source geometry, root state, and integrity counters are read back; authored intent alone is insufficient.
- F8.** Visual context has no collision API. Cameras and decoration may reveal the event but cannot change it.

Physical execution and evidence flow

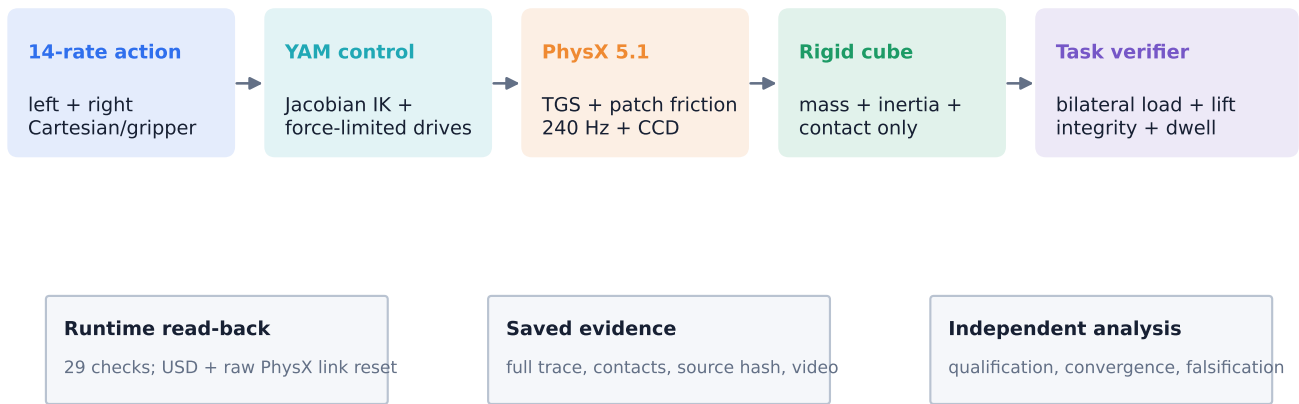


Figure 1: Physical execution and evidence flow. The 14-channel command drives both interface slots, but the reference sends zeros to the right slot. PhysX owns link, cube, and contact state; the task verifier consumes state but never applies force.

1.3 Claim hierarchy

Class	Examples	Permitted interpretation
Requested/source-exact	cube edge and spawn square; station dimensions and mount poses; vendored meshes, inertias, limits, and drive ceilings	definition or asset provenance, subject to runtime read-back
Runtime-audited	gravity, solver, friction model, CCD, mass, inertia, contact offsets, iterations, material bindings, GPU dynamics, physical root/link reset	statement about the executed Isaac stage
Engineering prior	density, friction, restitution, rigid-pad approximation, controller gains	selected starting value; never described as measured
Simulation outcome	lift, contact timing, normal impulse, trajectory, convergence, success, render	result of the exact source/settings; not hardware validation
Physical evidence	metrology, force/aperture curves, tilt/sled tests, tracked trajectories, held-out trials	absent until the calibration schema is populated from a named installation

2 Physical system and parameter provenance

2.1 Workcell geometry

The table top is centered at (0.600,0,0.745) m, with dimensions 595 mm×1,300 mm×10 mm. Its top surface is at 750 mm. The two arm bases are at (0.2525,+0.3100,0.7500) m and (0.2525,−0.3100,0.7500) m. These values are read from the vendored station description and checked against composed world bounds. The cabinet underneath and bolted arm-base interfaces are retained; only the two base/cabinet pairs that represent an intentional mounting overlap are filtered. Moving arm, finger, table, and cube pairs remain collidable.

The YAM asset contains two independent eight-DoF articulations: six revolute arm joints and two opposed prismatic finger joints per side. The source maximum efforts are 28 N m for joints 1–3, 10 N m for joints 4–6, and 8 N for each slide. Source MJCF gravity compensation on 22 named moving bodies is mapped to the same PhysX bodies. Link 1 and the physical D405 body on each arm retain ordinary gravity.

2.2 Cube mass and inertia

For edge a and density ρ , the cube model is

$$m = \rho a^3, \quad \mathbf{I}_C = \frac{ma^2}{6} \mathbf{I}_3. \quad (1)$$

The canonical prior uses $a = 0.050$ m and $\rho = 1,050$ kg m⁻³, giving $m = 0.13125$ kg and principal inertia $I = 5.46875 \times 10^{-5}$ kg m². Mass and inertia are authored explicitly and read back with a one-float32-ULP-scaled tolerance. That tolerance matters: an early high-density falsification correctly exposed that a fixed absolute tolerance could reject an exactly authored large float32 mass.

2.3 Parameter ledger

Quantity	Canonical value	Class	Qualification route
Cube edge	50 mm	requested	caliper/flatness/corner-radius survey
Cube density	1,050 kg m ⁻³	prior	mass plus three-axis metrology; ASTM D792 only for material coupon
Cube/table μ_s, μ_d	0.50, 0.40	prior	slow tilt plus multi-load/multi-speed sled
Finger/cube μ_s, μ_d	0.90, 0.75	prior	instrumented pad/cube sled and slip on-set
Restitution	0.02	prior	synchronized drop/rebound trials
Gravity magnitude	9.80665 m s ⁻²	source setting	runtime scene read-back; local gravimetry only for hardware claim
Physics/control rate	240 / 60 Hz	numerical	temporal convergence and timestamp audit
Solver / friction	TGS / patch	numerical	runtime token read-back and solver sensitivity
Cube iterations	32 position / 4 velocity	numerical	16/32/64 and 2/4/8 studies
Cube contact offset	0.30 mm	numerical	0.15/0.30/0.60 mm sensitivity
Robot contact offset	0.30 mm	numerical	runtime read-back on every finger collider
Success lift/dwell	50 mm / 0.20 s	task	pre-registered task definition; 200 mm falsification
Speed ceiling	0.15 m s ⁻¹	task	terminal trace verifier

The number of printed digits is an execution specification, not a statement of physical measurement precision. The calibration data format requires instrument uncertainty, environmental conditions, raw-file hashes, and frozen specimen splits before a coefficient may move from **PRIOR** to measured.

3 Governing rigid-body and contact mechanics

3.1 Free rigid body

With cube center $\mathbf{x}$, world linear velocity $\mathbf{v}$, angular velocity $\boldsymbol{\omega}$, world inertia $\mathbf{I}_W = \mathbf{R} \mathbf{I}_C \mathbf{R}^\top$, gravity $\mathbf{g}$, and contact wrenches $(\mathbf{f}_i, \mathbf{r}_i)$,

$$\dot{\mathbf{x}} = \mathbf{v}, \quad (2)$$

$$m \dot{\mathbf{v}} = m \mathbf{g} + \sum_i \mathbf{f}_i, \quad (3)$$

$$\mathbf{I}_W \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}_W \boldsymbol{\omega}) = \sum_i (\mathbf{r}_i - \mathbf{x}) \times \mathbf{f}_i. \quad (4)$$

PhysX advances a finite-step nonsmooth approximation; the equations state the modeled mechanics, not an assertion that the discrete solver is exact.

3.2 Unilateral contact and friction

For normal gap g_n , normal impulse λ_n , tangential impulse λ_t , and coefficient μ ,

$$g_n \geq 0, \quad \lambda_n \geq 0, \quad g_n \lambda_n = 0, \quad (5)$$

$$\|\lambda_t\|_2 \leq \mu \lambda_n. \quad (6)$$

This is the idealized Signorini–Coulomb statement underlying rigid contact and complementarity formulations.^[10,11] The installed engine uses TGS and patch friction, finite contact/rest offsets, iterative constraint solves, and GPU rigid dynamics.^[5–8] Static/kinetic friction coefficients are combined by the authored maximum rule, which is audited at runtime.

The callback provides contact points, normals, separations, and solved normal-impulse vectors. The reported per-step normal force is

$$F_{n,k} = \frac{1}{\Delta t} \sum_{p \in k} |\mathbf{J}_p \cdot \hat{\mathbf{n}}_p|. \quad (7)$$

Every point also stores the off-normal residual

$$\epsilon_{t,p} = \|\mathbf{J}_p - (\mathbf{J}_p \cdot \hat{\mathbf{n}}_p) \hat{\mathbf{n}}_p\|_2.$$

The residual audits the callback semantics; it is not a tangential-friction measurement. Internal patch-friction impulse is not exposed through this callback and is explicitly marked unavailable.

3.3 Task-direction grasp support

If two ideal opposed patches carry equal normal load N , a conservative vertical support condition is

$$2\mu_d N \geq mg, \quad N_{\min} = \frac{mg}{2\mu_d}. \quad (8)$$

For the canonical priors, $N_{\min} = 0.8581$ N per finger. The independent verifier recomputes this capacity from saved normal loads. This is only a gravity-direction capacity check. Two ideal point contacts do not establish full six-dimensional force closure; force/form closure and disturbance-wrench quality require a complete grasp matrix and admissible wrench set.^[9,12,13]

3.4 Finite-step numerical contract

The physics step is $\Delta t_p = 1/240$ s and the action interval is $\Delta t_c = 1/60$ s = $4\Delta t_p$. State is observed only after all four physics steps, while unsafe contact flags and peak loads accumulate over every substep. Rendering is never allowed to advance simulation during evidence capture. The numerical study estimates a local refinement difference

$$\delta_h(Q) = Q(\Delta t_p/2) - Q(\Delta t_p) \quad (9)$$

for terminal lift, duration, and peak contact load. A small δ_h supports local numerical stability near the chosen trajectory; it does not prove asymptotic order or global convergence for arbitrary contact sequences.

4 Robot dynamics, control, and exact reset

4.1 Physical articulation ownership

The robot is not a kinematic animation. PhysX integrates every link and joint under fixed bases, inertial properties, gravity/gravity compensation, collisions, drive gains, and effort limits. A normalized action

$$\mathbf{a} = [\mathbf{u}_L^v, \mathbf{u}_L^\omega, u_L^w, \mathbf{u}_R^v, \mathbf{u}_R^\omega, u_R^w] \in [-1, 1]^{14} \quad (10)$$

maps to bounded world-frame translational velocity, angular velocity, and aperture rate. Full scale is 0.18 m s^{-1} , 1.2 rad s^{-1} , and 0.04 m s^{-1} , respectively.

The command is a rate, not an accumulating hidden pose. At each action boundary, the backend forms a one-inverse-gain pose lookahead so the first proportional substep realizes the requested rate without accumulating target latency:

$$\mathbf{x}^* = \mathbf{x} + \frac{\mathbf{v}_{\text{cmd}}}{k_p}, \quad \mathbf{R}^* = \exp\left(\left[\frac{\boldsymbol{\omega}_{\text{cmd}}}{k_R}\right]_{\times}\right) \mathbf{R}. \quad (11)$$

4.2 Jacobian controller

Let $\mathbf{e}_x = \mathbf{x}^* - \mathbf{x}$ and $\mathbf{e}_R = \log(\mathbf{R}^* \mathbf{R}^\top)^\vee$. The desired twist is clipped componentwise by physical speed limits:

$$\boldsymbol{\xi}^* = \begin{bmatrix} \text{clip}_{v_{\max}}(k_p \mathbf{e}_x) \\ \text{clip}_{\omega_{\max}}(k_R \mathbf{e}_R) \end{bmatrix}. \quad (12)$$

The controller uses the measured PhysX spatial Jacobian, shifted from the link center of mass to the authored grasp site. Damped least squares gives

$$\dot{\mathbf{q}} = \mathbf{J}^\top (\mathbf{J} \mathbf{J}^\top + \lambda^2 \mathbf{I}_6)^{-1} \boldsymbol{\xi}^*, \quad \mathbf{q}^+ = \text{clip}_{[\mathbf{q}_{\min}, \mathbf{q}_{\max}]}(\mathbf{q} + \Delta t_p \dot{\mathbf{q}}). \quad (13)$$

The articulation controller then applies source-bounded position drives. In conceptual form,

$$\tau_j = \text{sat}_{\tau_{j,\max}} (k_{p,j}(q_j^+ - q_j) - k_{d,j}\dot{q}_j). \quad (14)$$

No operational-space wrench is applied directly to the cube.

4.3 First divergent state and reset repair

The first campaign initially appeared to compare identical states because joint positions, joint velocities, cube pose, seed, simulated warm-up time, policy, and commands matched. Its canonicalized initial-state hashes nevertheless differed. The first divergent fields were the saved grasp transforms, at sub-25 μm scale. A matched trace isolated the mechanism:

- R1.** After a fixed simulated robot warm-up, generalized coordinates were reauthored exactly.
- R2.** The USD grasp-site world transform remained stale until an application update and still differed by solver/-timestep history.
- R3.** PhysX link transforms were much closer to source forward kinematics, but root/link velocities retained warm-up-dependent residuals when only joint state was reset.
- R4.** Explicitly reauthoring the fixed root pose and setting root velocity to zero, followed by one atomic eight-coordinate state write, made every raw link transform and velocity bit-exact across matched 120 Hz and 240 Hz cells.
- R5.** The controller now reads the PhysX link-6 tensor transform and composes it with the source-authored link-6-to-grasp transform. The verifier independently reconstructs the recorded grasp pose from those raw values.

This was a correctness bug, not an “expected nondeterminism” or an architectural ceiling. The failed campaign was rejected; no tolerance was widened to make it pass. The production invariant records both arms’ complete raw link state in every initial-state artifact and requires zero link velocity.

4.4 Matched boundary-failure isolation

Seed 588 exposed a separate reference-policy regression at the far $+x/+y$ boundary: both fingers made force-bearing contact, but the cube was lost instead of lifted. The probe changed only the pre-close Cartesian position tolerance. Seed, sampled origin, complete physical reset state, specification, every other policy field, backend process, GPU, and execution order were held fixed. All four cells recorded exact initial-state equality and passed the 29 runtime checks.

Gate (mm)	Error at close (mm)	Peak lift (mm)	Peak finger B (N)	Result
4.0	3.951	0.553	8.637	fail
2.0	1.975	0.640	9.808	fail
1.0	0.995	83.304	12.344	pass
0.5	0.469	83.419	21.040	pass

Table 1: Matched seed-588 phase-gate sweep. “Lift” is measured from the immutable sampled origin; pass still requires the complete history-sensitive task predicate.

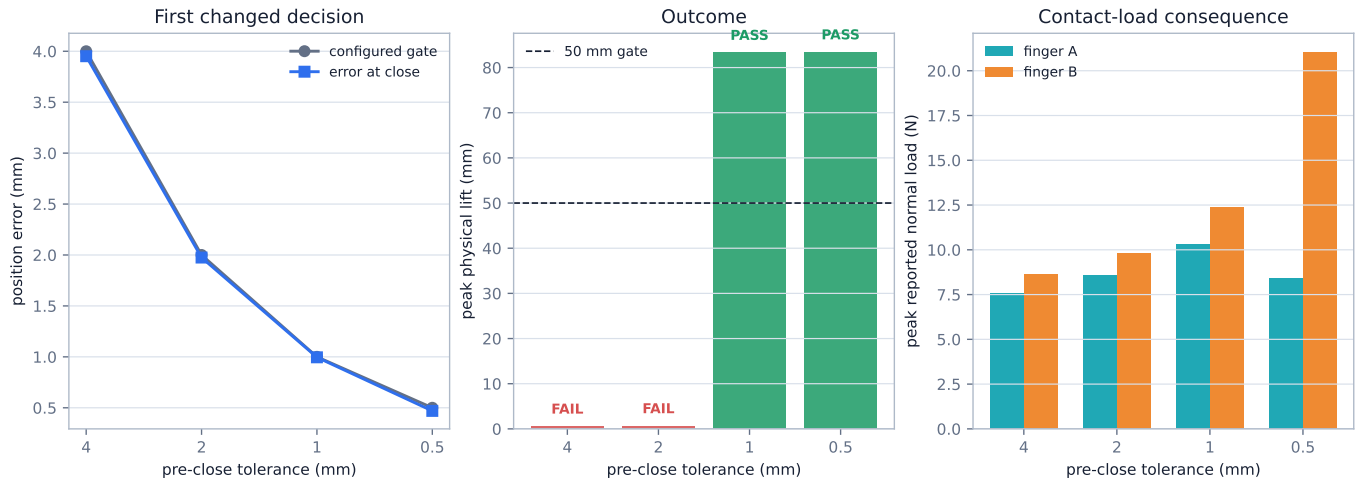


Figure 2: The first intentionally changed state is the close-transition time and its physical pose error. The looser gates close early and later lose the cube; the 1 mm and 0.5 mm gates lift it. The tighter 0.5 mm cell raises peak finger-B load to 21.040 N, so it is not adopted merely because it also succeeds.

Fact. The 4 mm and 2 mm cells enter closure at 3.951 mm and 1.975 mm error, obtain bilateral contact, and reach only 0.553 mm and 0.640 mm peak lift. The 1 mm and 0.5 mm cells enter closure later and reach 83.304 mm and 83.419 mm.

Conclusion. Premature close-phase entry is the concrete mechanism for this boundary failure in the matched reference-policy probe. The 1 mm gate is the least restrictive tested successful setting and has substantially lower finger-B peak load than the 0.5 mm cell. It is therefore retained and is subsequently tested, without per-seed changes, in the complete 25-placement qualification. This conclusion is about the demonstration policy; it does not reinterpret failed pickup attempts as a limitation of PhysX or of the environment.

5 Scene construction, collision, and rendering

5.1 Mechanical scene

The cube is a single dynamic USD cube with collision, mass, diagonal inertia, CCD, zero sleep/stabilization thresholds, 32/4 solver iterations, and the declared contact offsets. The table and cabinet are finite colliders. Finger materials are bound to every finger collision shape and audited by path. Contact reporting is enabled on the cube and relevant arm bodies, and the monitor maps contact pairs to left finger A/B, right arm, table, and station classes.

The imported station was generated by a newer Isaac toolchain than the checked Isaac 5.1 runtime. Two compatibility transformations are representation-only and audited: nested rigid bodies receive world-transform-preserving reset stacks, and four source-massless TCP/camera coordinate-frame bodies lose arbitrary rigid-body/mass APIs while keeping their authored transforms. The audit requires zero initial transform change for both operations and exactly two eight-DoF articulations afterward.

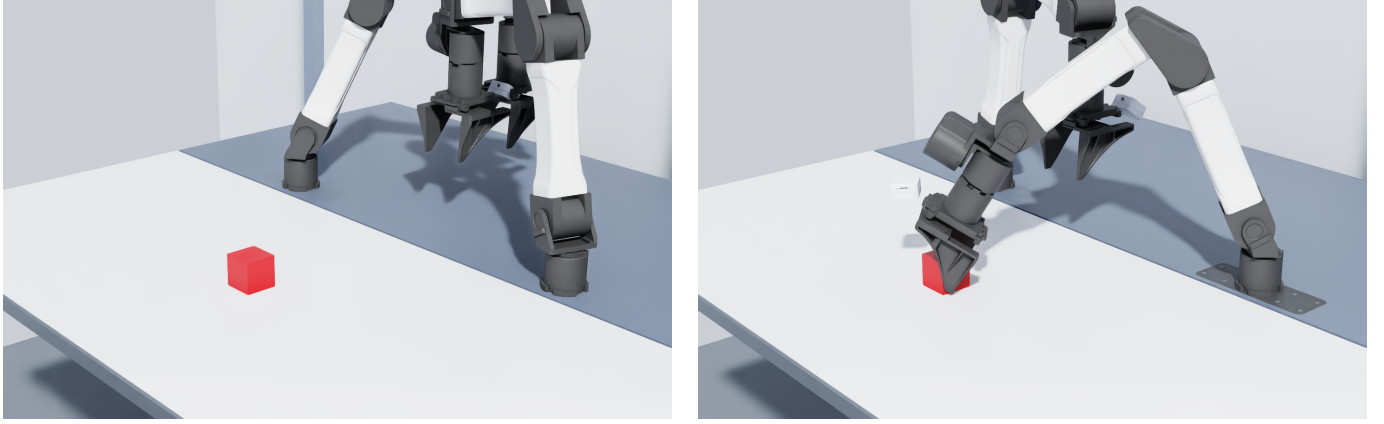
5.2 Professional visual scene without physical contamination

The render presents the real YAM visual meshes, white work table, dark cabinet, trim, cabinet doors and handles, a back wall, overhead and warm fill lighting, and a saturated red cube. These additions make scale and workcell context legible. They are explicitly noncolliding visual context; the runtime audit checks that each exists and lacks collision API.

Three calibrated-in-software evidence viewpoints are authored:

- **task:** oblique close view of grasp and lift;
- **overview:** complete bimanual station and workspace;
- **contact:** low close view of finger/cube/table separation.

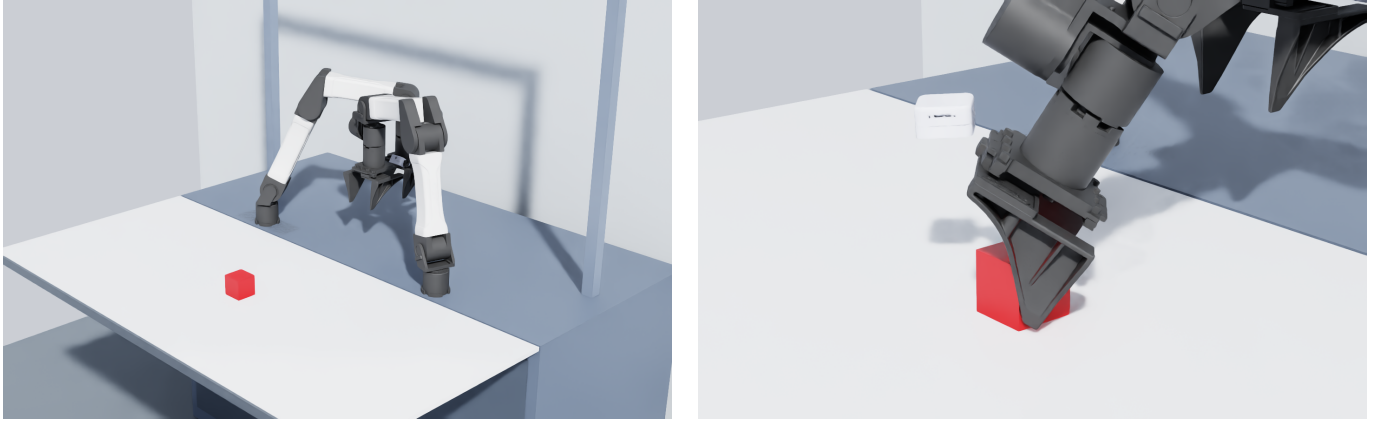
Camera existence and positive focal length are read back. Capture calls temporarily render without advancing physics. Each PNG records resolution, simulation timestamp, and SHA-256. Fixed-time video frames are selected by tag, encoded to H.264/yuv420p at a rate derived from simulation timestamps, then decoded independently to verify frame count, resolution, and frame rate.



(a) Task camera before contact.

(b) Force-supported table separation.

Figure 3: Visual evidence uses the same physical execution source as qualification and numerical evidence. No visual shell drives the cube.



(a) Overview: both YAM arms, finite table, cabinet, and workcell context.

(b) Contact view: finite finger geometry at closure.

Figure 4: Independent evidence cameras expose whole-scene non-assistance and local contact. Decorative wall/cabinet details are audited as noncolliding.

6 Task interface, reward, and verifier

6.1 Observation contract

One control-boundary observation contains cube pose and six-dimensional velocity; both grasp-site poses; both apertures; left-finger contact booleans and resultant/normal loads; right-arm/cube, cube/table, and left-arm/table contact; sampled origin; reset-pose-write count; and attachment count. Positions are SI metres, velocities SI units, quaternions scalar-first, and all public arrays have fixed shape.

The right arm is not deleted or frozen. It remains physically simulated, which makes the non-assistance claim falsifiable. The reference policy writes exact zeros to action indices 7–13; both the task and runner track the maximum absolute right action over the complete episode.

6.2 Dense potential is not success

Let d be grasp-to-cube distance, $c_A, c_B \in \{0, 1\}$ the two left contact indicators, h lift from the sampled origin, and $h_* = 50$ mm. The diagnostic potential is

$$\Phi(s) = 0.35e^{-d/0.10} + 0.30\frac{c_A + c_B}{2} + 0.35 \operatorname{clip}\left(\frac{h}{h_*}, 0, 1\right). \quad (15)$$

The step reward is

$$r_t = \Phi(s_{t+1}) - \Phi(s_t) + 10\mathbb{I}_{\text{success}} - 10\mathbb{I}_{\text{integrity/contact failure}}. \quad (16)$$

Reward is computed after physics and never applies force, changes contact, or writes pose.

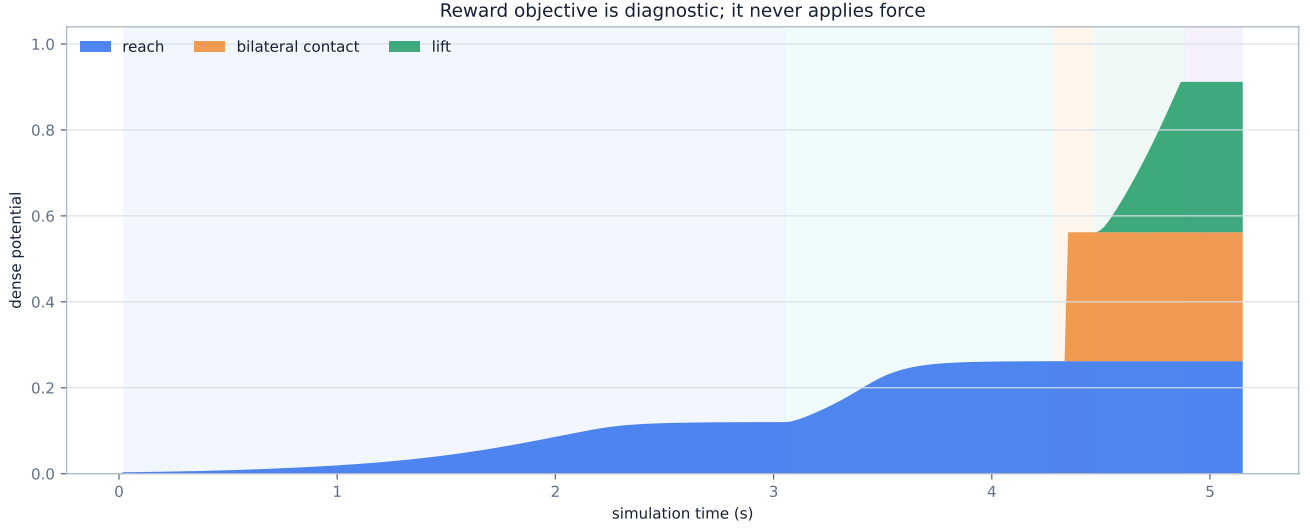


Figure 5: Potential components along the retained reference episode. Sparse success remains authoritative; the plot is explanatory.

6.3 History-sensitive success

For continuous dwell interval $\mathcal{D}$ of duration at least $T_* = 0.20$ s, success is

$$\begin{aligned} \mathcal{S} = & [\forall t \in \mathcal{D} : c_A(t) \wedge c_B(t)] \wedge \left[\min_{t \in \mathcal{D}} h(t) \geq h_* \right] \\ & \wedge \left[\max_{t \in \mathcal{D}} \|\mathbf{v}_C(t)\| \leq 0.15 \text{ m s}^{-1} \right] \wedge \neg c_{C,T} \wedge \neg c_{L,T} \wedge \neg c_{R,C} \\ & \wedge [N_{\text{pose}} = 1] \wedge [N_{\text{attachment}} = 0]. \end{aligned} \quad (17)$$

At least one bilateral force-bearing frame must precede lift. A cube displacement above 35 mm in one 60 Hz interval is an integrity failure named *excessive control-step displacement*. The name deliberately does not invent a cause: in the zero-friction falsification, the cube leaves the tabletop and the large displacement is a physical gravity-driven fall, not a teleport.

7 Numerical evidence campaign

7.1 Pre-registered matched design

The campaign manifest fixes seed 3819, the sampled cube origin, 60 Hz controller, reference policy, GPU device, source hash, 0.20 s simulated robot warm-up, exact post-warm-up physical root/generalized state, and task predicate. Every cell runs in a fresh Isaac process. The manifest records the complete set of intentionally changed fields. A comparison is inadmissible if initial state, simulated waiting time, execution order, commands, or source differs without being named.

Family	Cells	Changed variable	Question
Repeatability	3	none	Does a fresh process reproduce the canonical trajectory exactly?
Temporal	3	120/240/480 Hz	Does the result remain stable at 2/4/8 physics substeps per action?
Position solve	3	16/32/64 iterations	Is the contact-supported outcome stable around the selected TGS position count?

Family	Cells	Changed variable	Question
Velocity solve	3	2/4/8 iterations	Is restitution/friction response sensitive to velocity iteration count?
Contact offset	3	0.15/0.30/0.60 mm	Does contact generation distance materially alter the episode?
Density	3	840/1050/1260 kg m ⁻³	Does a $\pm 20\%$ mass/inertia perturbation change the task result?
Finger friction	3	canonical $\times \{0.8, 1, 1.2\}$	Does moderate pad/cube friction uncertainty alter success?
Restitution	3	0/0.02/0.10	Is the low-speed pickup driven by rebound choice?
Solver/CCD	3	TGS/PGS; CCD on/off	Are algorithm and tunneling settings active outcome sensitivities?
Physical falsification	2	all friction zero; density $\times 16$	Does the task fail when necessary support mechanisms are removed or capacity is exceeded?
Verifier falsification	1	required lift 200 mm	Does a physically good but insufficient lift remain a failure?

Although the table reuses the canonical cell across triplets, the executable manifest contains 22 unique process cells. The summary accepts only one execution-source hash and one canonical initial-state hash. It records per-cell artifact hash, full trajectory hash, all overrides, runtime-audit failures, terminal state, lift, duration, speed, contact loads, and baseline deltas.

7.2 Repeatability and exact initial state

The three canonical processes share one raw initial-state fingerprint and one complete trajectory fingerprint. “Initial state” includes cube pose/velocity, both eight-coordinate articulation states, both grasp poses, all 28 physical link transforms, and all 28 physical link velocities. The final summary reports 1 unique initial state across all 22 cells. This gate is deliberately stronger than comparing random seeds or joint arrays.

The campaign does not claim that GPU contact is universally deterministic. It reports exact reproduction for the declared machine, driver/runtime, source, and episode. Repeatability is an observed local property, while initial-state equality is a controlled fact.

7.3 Temporal and solver convergence

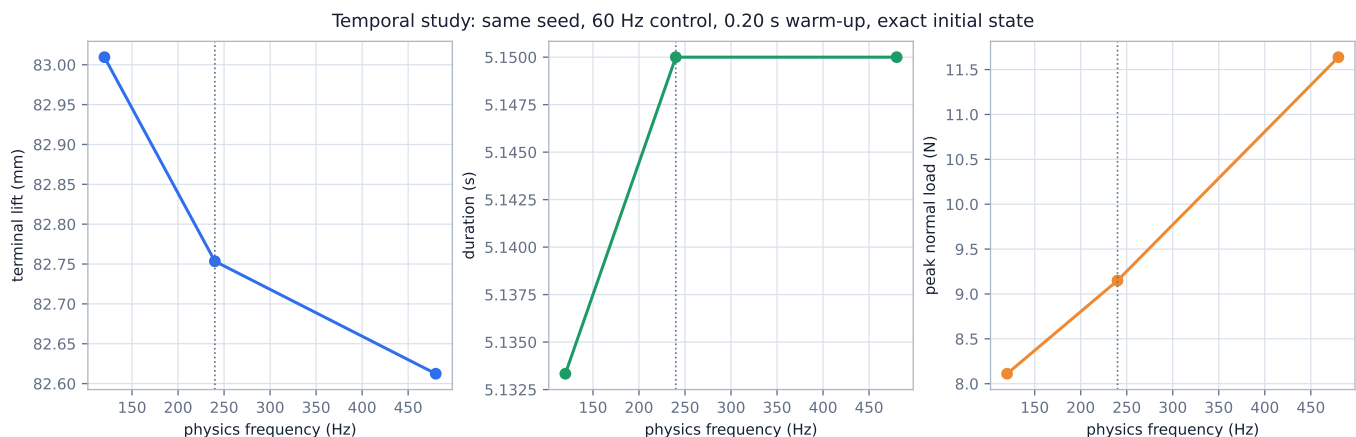


Figure 6: Matched temporal study. The canonical 240 Hz result is compared with 120 and 480 Hz while control remains at 60 Hz. Initial physical state, seed, policy, and task gate are exact.

Terminal lift is 83.010 mm, 82.753 mm, and 82.612 mm at 120, 240, and 480 Hz. The report retains duration and peak-load shifts as well as terminal lift; a terminal scalar alone can hide timing or impulse changes. These three cells support local step refinement around the selected action, not a claim that every collision-heavy YAM trajectory is converged.

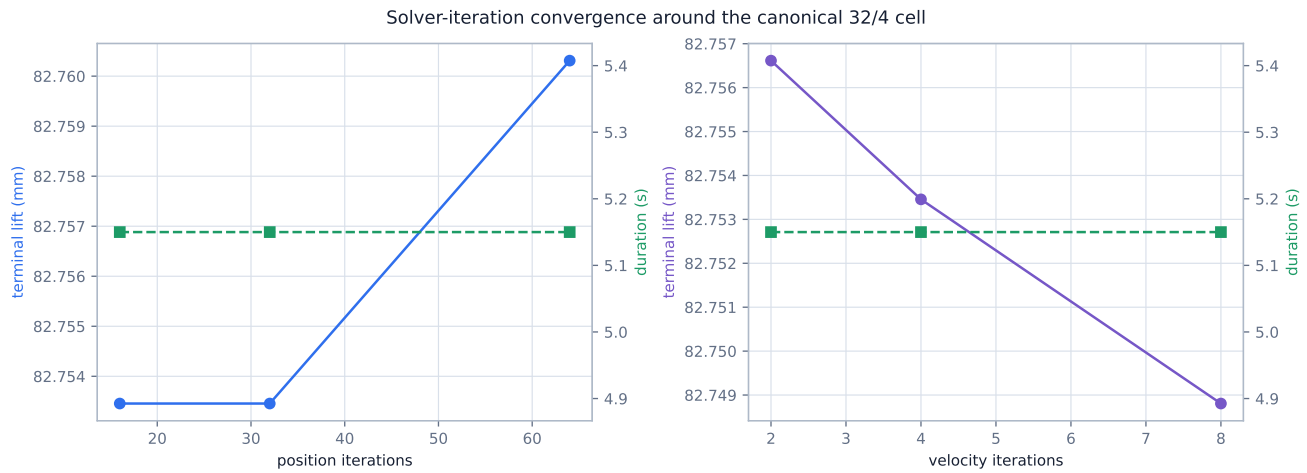


Figure 7: Position- and velocity-iteration studies around the canonical 32/4 profile. Each point is a fresh matched process.

The iteration study is interpreted similarly. If terminal lift remains close but peak normal force changes, the contact load remains numerically sensitive and must be included in calibration uncertainty. The runtime audit proves the requested iteration count read back on the cube in every cell; labels are not inferred from filenames.

7.4 Material, contact, and algorithm sensitivity

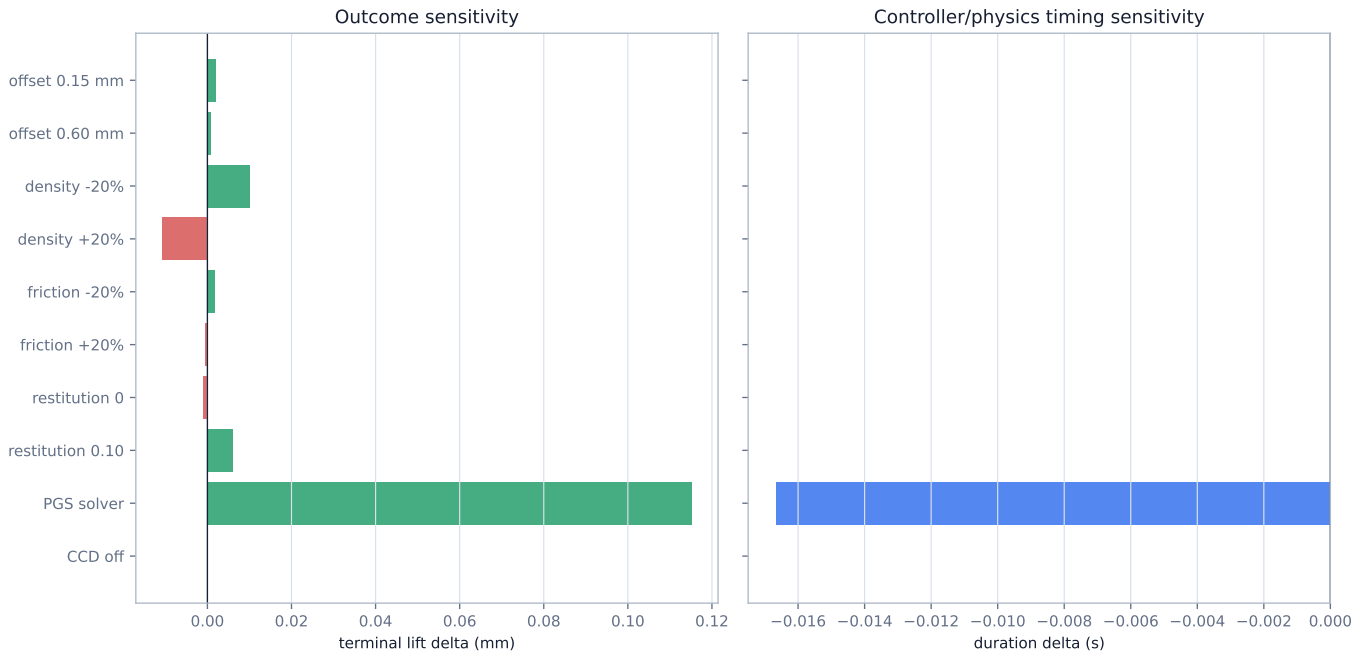


Figure 8: Signed outcome and duration deltas for contact offset, density, friction, restitution, solver choice, and CCD. The largest non-falsification terminal-lift deviation is 0.2561 mm.

Sensitivity does not validate the priors. It identifies which conclusions survive modest perturbations and which measurements deserve priority. Density changes mass and all principal inertias together. Finger friction changes both pad coefficients while leaving cube/table contact fixed. Contact offset changes collision generation distance, not solid dimensions or rest offset. PGS changes the solver while preserving frequency, iterations, geometry, and commands. Disabling CCD tests this low-speed episode, not whether CCD is globally unnecessary.

7.5 Retained falsifications

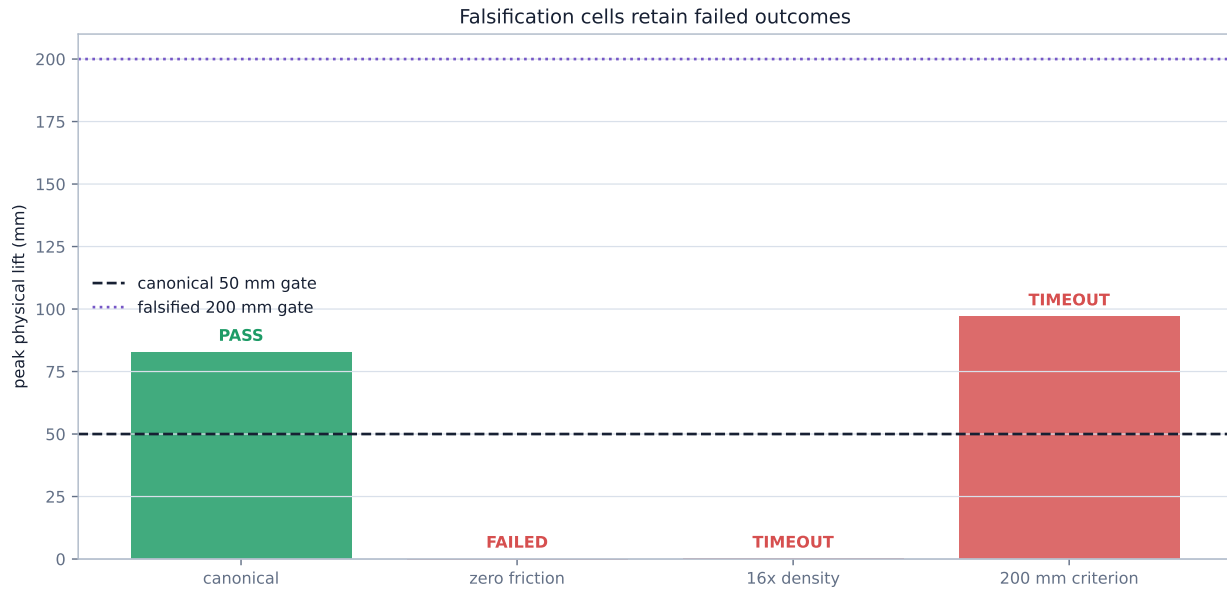


Figure 9: Failed outcomes are retained rather than discarded. Green denotes canonical success; red denotes a physical or predicate failure.

All friction zero. Setting only finger coefficients to zero would not remove friction because the authored combine rule is maximum and the cube material would still contribute. The corrected cell sets cube, finger, and table static/dynamic coefficients to zero. The fingers push the cube beyond the finite tabletop during closure; after contact loss it free-falls. The first displacement above the control-boundary guard occurs with no contact and increasing downward speed. Peak lift is 0.000 mm; the episode fails rather than being cosmetically rescued.

Sixteen-times density. An earlier eight-times cell still succeeded because measured normal loads and the selected friction prior retained sufficient support capacity. That is evidence against calling eight-times mass an adverse test. The final cell uses $\rho = 16,800 \text{ kg m}^{-3}$, giving $m = 2.10 \text{ kg}$. Bilateral contact occurs and the controller enters lift, but peak cube lift is only 0.0411 mm; it times out on the table.

200 mm criterion. The physical trajectory reaches 96.945 mm, but the frozen success predicate requires 200 mm; the episode times out. This proves the verifier does not equate “grasped” or controller phase with task success.

Rejected inference. A failed falsification is not evidence that pickup is impossible, that PhysX has an architectural ceiling, or that the task should be relaxed. It is evidence about one deliberately changed mechanism. The report names the first divergent physical state and preserves every changed variable.

8 Reference episode and motion evidence

8.1 Observable-state reference policy

The policy follows approach, tool-axis descent, closure, tool-axis lift, and hold phases. It uses only the public observation and immutable sampled origin. The desired wrist is tilted 32° toward the cube because a vertical wrist was outside the exact YAM workspace at the far $+x/-y$ corner. A five-point inverse-kinematics probe found the tilted pose reachable at all four near-corner samples and center.

The imported grasp marker is not the physical center of the long inner pads in this tilted configuration. A collider measurement locates the pinch midpoint 25.6 mm radially inward of the marker. The commanded marker is therefore shifted outward by the same amount and placed 13.5 mm above the cube center. A matched height sweep found 12.5 mm to be the first sampled collision-free value; the retained target adds 1 mm clearance. Closure begins only after the measured grasp site is within 1 mm of that target, as selected by the matched seed-588 gate sweep above. These are policy geometry and transition values, not changes to cube or finger colliders.

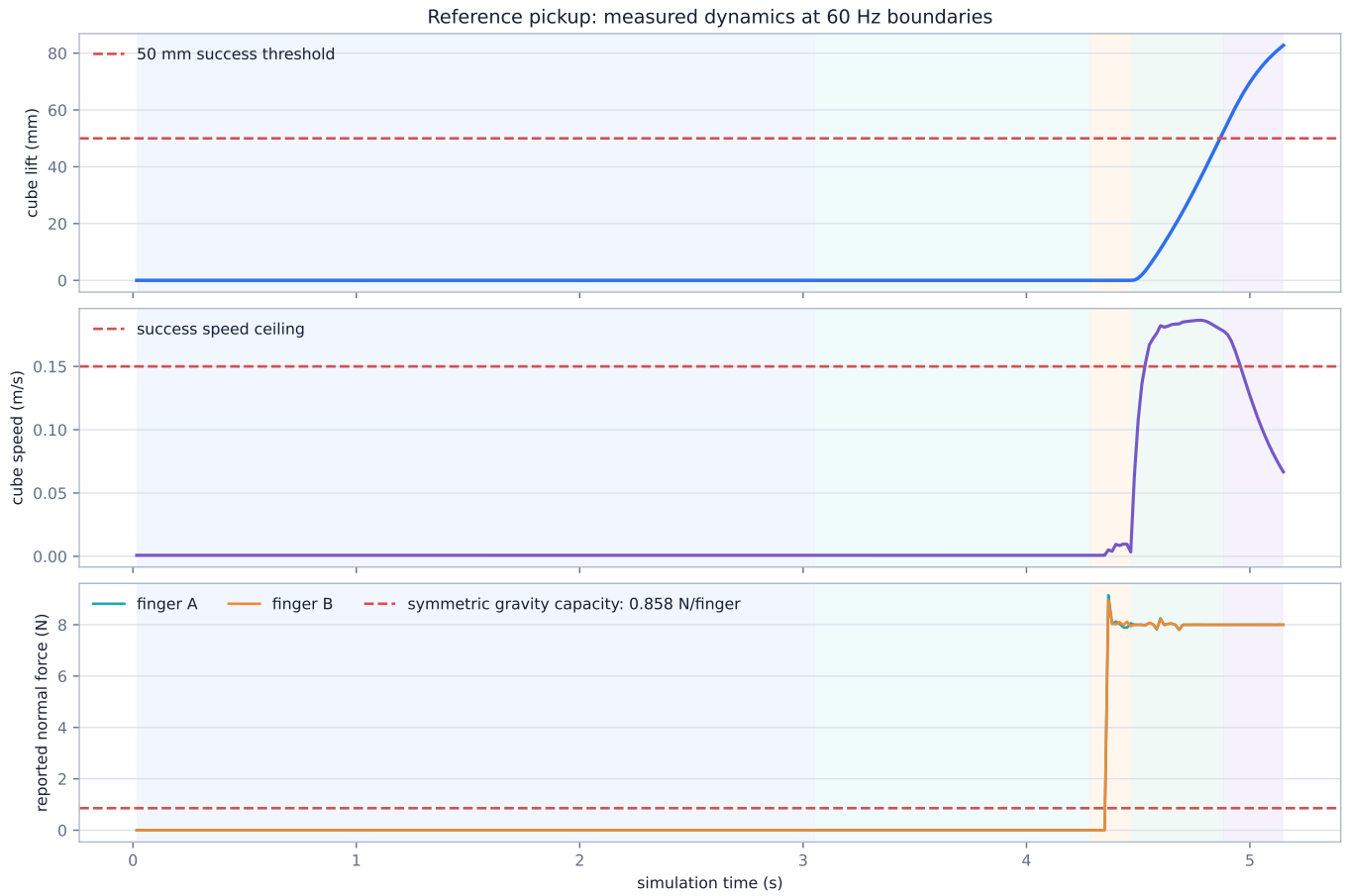


Figure 10: Seed 3819 reference dynamics. Lift, speed, and both reported normal loads are plotted at every saved 60 Hz boundary; phase shading comes from controller state.

The retained episode completes in 5.1500 s, reaches 82.753 mm terminal lift, and succeeds at cube speed $0.066658 \text{ m s}^{-1}$. Peak reported normal loads are 9.1492 N and 8.9495 N. These are simulation values obtained from Eq. (7); they are not fingertip-sensor measurements.

8.2 Controller tracking

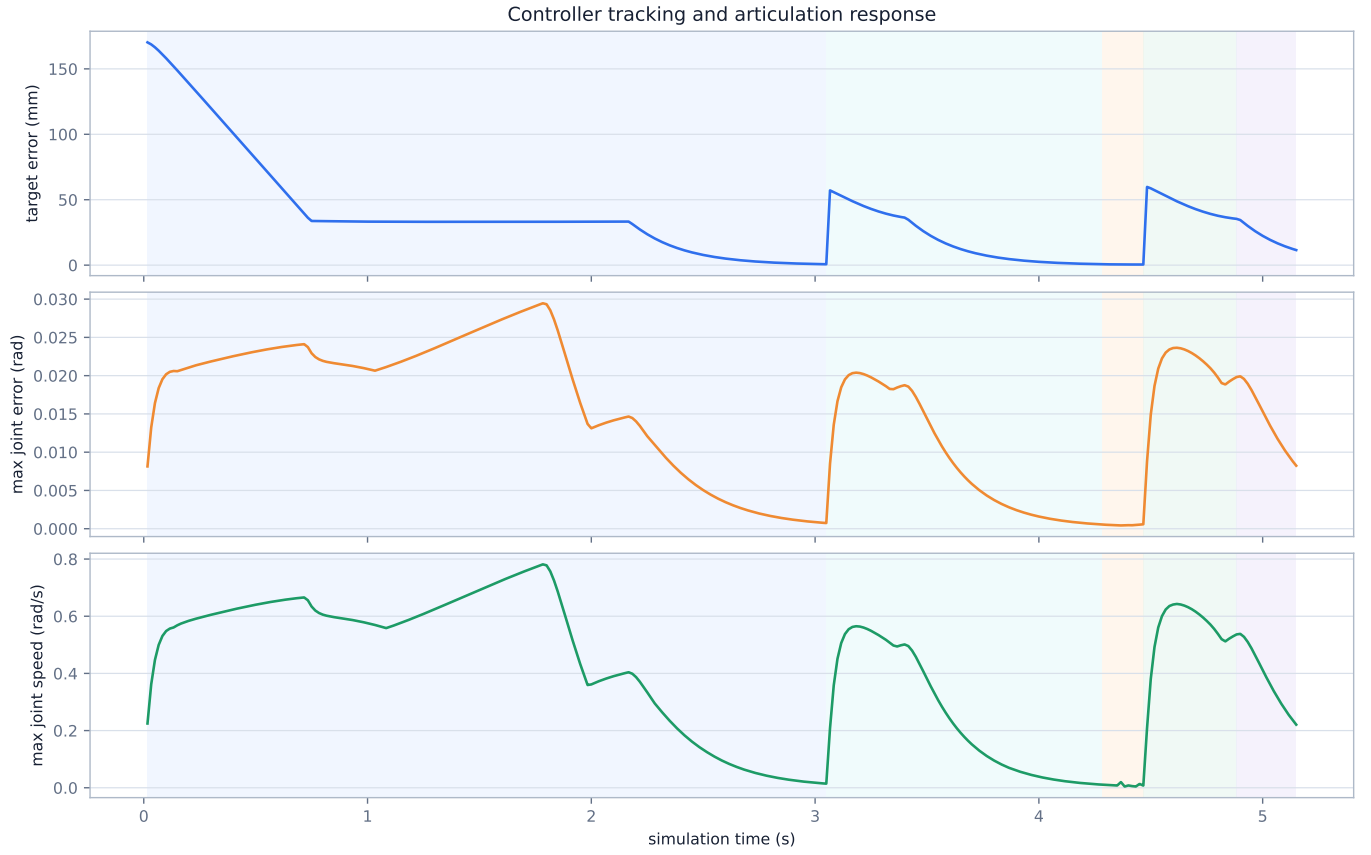


Figure 11: Measured grasp-target error, maximum joint-target error, and maximum left joint speed. The plot exposes articulation tracking rather than assuming command equals physical pose.

Each trajectory sample stores current joint state, current grasp pose, controller joint and Cartesian targets, action, cube state, contact state, and task phase. This exposes drive lag, saturation, and phase-transition dependence.

8.3 Six motion frames

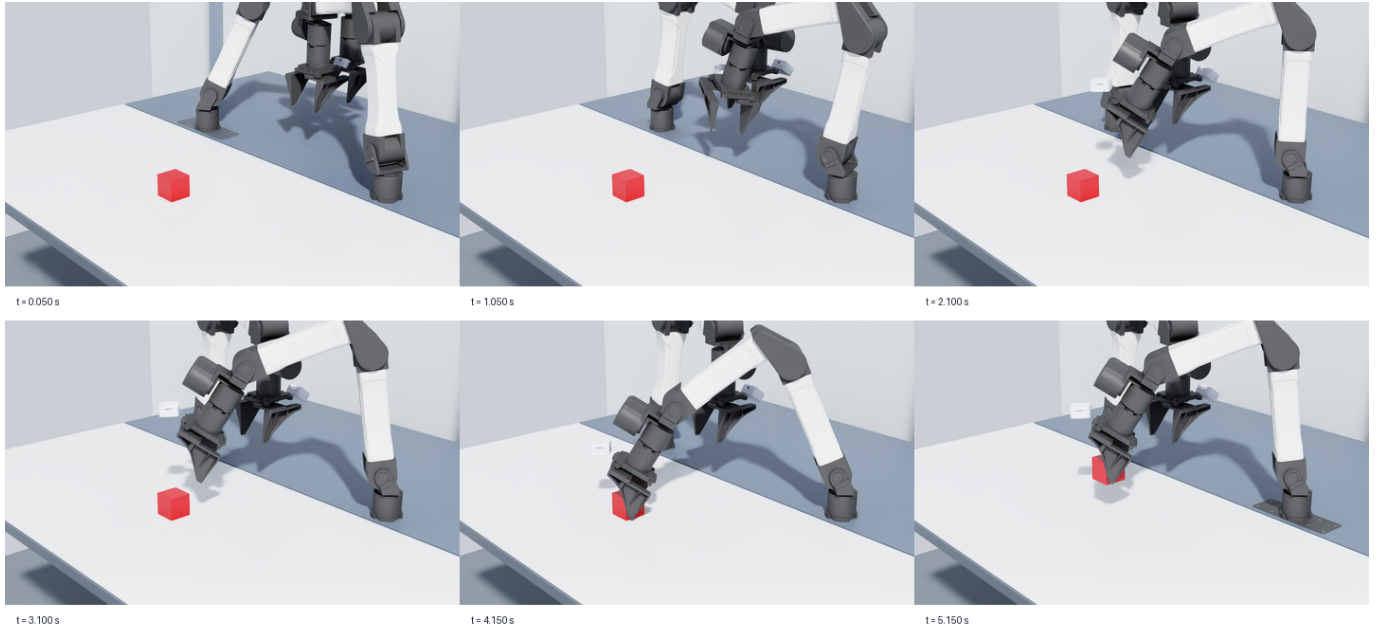


Figure 12: Fixed-simulation-time frames sampled from the encoded task-camera sequence. The physical trace, PNGs, MP4, and decode audit are separate hash-bound artifacts.

9 Frozen qualification

9.1 Boundary and random placements

Qualification uses two pre-registered panels under one final execution source:

- four near-corner samples plus center, chosen to exercise workspace and approach geometry;
- 20 seeds fixed before the final run, sampled uniformly over the requested square.

All 25/25 pass the same history-sensitive predicate. Sampled coordinates span $x \in [0.501597, 0.699692]$ m and $y \in [-0.096769, 0.098802]$ m. Terminal lift spans 80.414 mm–85.126 mm; duration spans 4.6000 s–6.2833 s; success speed spans 0.063738 m/s–0.072452 m/s; and the larger of the two per-episode peak normal loads spans 8.1130 N–13.1467 N.

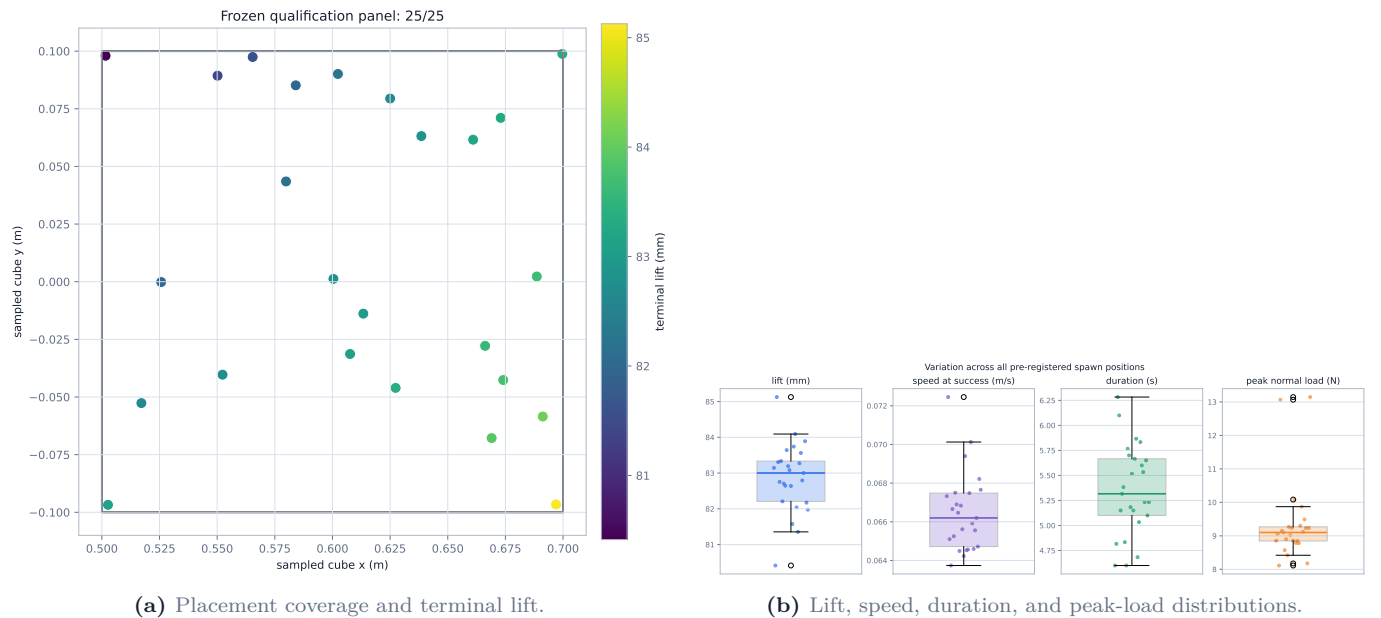


Figure 13: The qualification panel is broader than the single reference video and is not selected by outcome.

9.2 Independent artifact verifier

The verifier does not trust top-level `pass`. It independently checks schema, run kind, spec, execution hash, seed panel, success state, lift, dwell, speed, prohibited contacts, integrity counters, complete trajectory actions, right-arm zeros, runtime checks, and gravity-direction support capacity. It reconstructs each initial grasp pose from the raw PhysX link-6 transform and recorded authored local transform, validates canonical roots, normalized link quaternions, zero link velocities, exact home coordinates, and terminal observation/time consistency.

The boundary and random artifacts are accepted only if they share one execution source. Visual evidence and the numerical campaign must share that same source before the figure generator emits this report's macros. This prevents a favorable older qualification or prettier older video from being silently mixed with revised physics.

10 Runtime audit and provenance

10.1 Twenty-nine named runtime checks

Audit gate	Concrete read-back
Two YAM articulations / eight DoF each	articulation-root paths, count, names, joint order, DoF count
Cube exact geometry, mass, inertia	USD size, MassAPI, diagonal inertia, analytical expectation and ULP tolerance
Cube dynamics and collision	RigidBodyAPI and CollisionAPI presence
Table collision and source bounds	CollisionAPI plus composed world AABB
Cabinet/source bases	composed bounds and both arm-base positions
Physics scene configuration	gravity vector/magnitude, solver token, patch friction, CCD, GPU dynamics, GPU broadphase
Cube solver/contact configuration	contact/rest offsets, CCD, sleep/stabilization thresholds, 32/4 iteration read-back
Table/finger contact configuration	offsets/rest offsets for table and every finger collider
Material configuration	static/dynamic coefficients, restitution, combine modes, and every binding target
Visual context noncollision	context prim exists and lacks CollisionAPI
Three evidence cameras	prim exists and positive focal length
Bolted mount filtering	exactly the two declared base/cabinet relationships
Canonical home state	six arm coordinates and aperture per arm
Canonical physical reset	raw root transform error; every link linear/angular velocity exactly zero
Physical grasp provenance	PhysX link transform composed with authored local marker transform
Source gravity compensation	exact 22-body set and four uncompensated dynamic names
Cube visual, spawn, integrity	red display color, origin in square, one reset write, zero attachment
Finger binding/contact reports	positive bound-collider count and enabled left-arm reporting

Several rows contain multiple Boolean checks; the executable dictionary contains 29 named Booleans. The audit also records full values, not only Booleans, so a reviewer can detect an overbroad tolerance or wrong expected value.

10.2 Source identity

The execution manifest hashes physical task/runtime files and the complete YAM mechanical asset. Evidence software—verifier, plotting, report, and tests—has a separate hash so editorial changes cannot masquerade as changed physics. Conversely, any change to specification, backend, task, controller, policy, runner semantics, or mechanical asset changes the execution hash and invalidates final evidence.

Each file digest is SHA-256. The combined digest is computed from sorted path/digest pairs with an unambiguous delimiter. Artifact, PNG, MP4, figure, and PDF hashes are recorded in their respective manifests.

The final inventory contains two evidence-software hashes. The second begins when the verifier CLI gains an explicit visual-run-kind option, before the overview and contact traces are generated; the physical execution hash is unchanged.

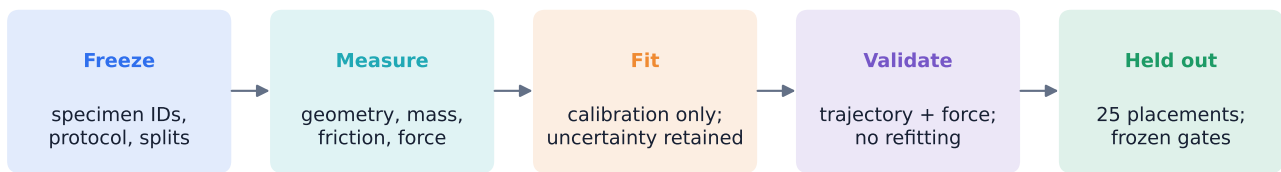
Qualification and all three visual artifacts are subsequently rechecked with the current verifier. This distinction preserves the physical identity claim without concealing an evidence-tool revision.

11 Physical calibration and held-out validation

11.1 Why calibration is still open

A numerically repeatable simulator can be precisely wrong. The current coefficients were selected as plausible priors before any named cube, pad, table, robot, temperature, humidity, wear state, pose tracker, or force sensor was supplied. Accordingly, the yellow calibration label appears on the cover and every page. A successful video, a good 25/25 simulation panel, and small timestep deltas do not remove it.

Physical calibration path — not yet executed



Raw data are immutable and SHA-256 identified. Simulation traces are never accepted as physical measurements.

Figure 14: Pre-registered calibration and validation flow. Raw physical data are immutable and never replaced by simulator traces.

11.2 Machine-readable measurement contract

`calibration/schema.json` requires:

- immutable installation, arm, gripper-pad, table, cube, specimen-lot, and split IDs;
- instrument model, serial number, calibration certificate/date, uncertainty, sampling rate, and coordinate frame;
- protocol version, operator, temperature, relative humidity, cleaning, wear conditioning, and approach direction;
- raw artifact path and SHA-256, with transformations stored only as separate derived artifacts;
- geometry, mass, center of mass, friction, restitution, aperture/load, pose/joint tracking, and held-out task records;
- explicit exclusions and their pre-computation reasons.

The synchronized CSV template includes time, commanded and measured joints, commanded and measured end-effector pose, aperture, bilateral normal loads, cube pose/twist, contact flags, environmental data, and quality flags. Empty templates are retained rather than filled with simulated values.

11.3 Identification experiments

Geometry and inertia. Measure each cube edge on three axes, face flatness, corner radius, mass, and center of mass. For a measured rectangular cuboid (a, b, c) rather than an assumed cube,

$$\mathbf{I}_C = \frac{m}{12} \text{diag}(b^2 + c^2, a^2 + c^2, a^2 + b^2). \quad (18)$$

Survey table and arm bases into one workcell frame, including table level and local compliance.

Friction. Run a slow tilt test for slip onset,

$$\hat{\mu}_s = \tan \theta_{\text{slip}}, \quad (19)$$

then constant-speed sled trials over multiple normal loads, material directions, and speeds,

$$\hat{\mu}_d(v, N) = \text{median}_{t \in \mathcal{W}_{\text{steady}}} \frac{|F_t(t)|}{F_n(t)}. \quad (20)$$

Preserve the distribution and trial-level exclusions. ASTM D1894 supplies useful terminology and apparatus guidance, but any adaptation to a rigid cube/finger pad must be declared.^[18]

Restitution. From synchronized pre-impact and rebound normal velocities,

$$\hat{e} = -\frac{v_n^+}{v_n^-}, \quad v_n^- < 0, \quad (21)$$

over multiple impact speeds and orientations. Do not infer restitution from a visually pleasing bounce.

Gripper and robot. Measure aperture versus slide state, bilateral normal load versus command/aperture/speed, six-dimensional end-effector accuracy, repeatability from multiple approach directions, and joint tracking. ISO 9283 and ISO/TR 13309 provide the accuracy/repeatability and metrology framework; NIST hand/grasp work motivates reproducible load, grasp, and task measures.^[14–17]

11.4 Fit, validation, and held-out acceptance

Let physical samples be $\mathbf{y}_i$, time-aligned simulator outputs $\hat{\mathbf{y}}_i(\boldsymbol{\theta})$, instrument covariance $\boldsymbol{\Sigma}_i$, and regularization centered on independently measured priors $\boldsymbol{\theta}_0$. Calibration minimizes

$$\mathcal{L}_{\text{cal}}(\boldsymbol{\theta}) = \sum_{i \in \mathcal{C}} (\mathbf{y}_i - \hat{\mathbf{y}}_i)^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{y}_i - \hat{\mathbf{y}}_i) + \lambda \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|_2^2. \quad (22)$$

Validation and held-out splits are never refit. Acceptance must use frozen bounds for trajectory error, contact timing, peak and impulse load, slip, terminal pose, and success rate:

$$\text{RMSE}_x = \sqrt{\frac{1}{N} \sum_i \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2}, \quad \Delta t_c = |t_c - \hat{t}_c|. \quad (23)$$

At minimum, the held-out task panel repeats the four near-corners, center, and 20 pre-registered random positions with cubes and pad conditions excluded from calibration.

12 Related work and design decisions

12.1 Rigid contact and grasping

Stewart–Trinkle and Anitescu–Potra provide foundational nonsmooth/complementarity formulations for frictional rigid-body time stepping.^[10,11] Modern Robotics, Bicchi, and Ferrari–Canny distinguish form closure, force closure, and disturbance-wrench grasp quality.^[9,12,13] This environment uses those distinctions to avoid overclaiming: bilateral normal load plus Eq. (8) is a task-direction capacity audit, not a six-dimensional closure certificate.

PhysX documentation defines the installed TGS, iteration, friction-patch, collision-offset, contact-report, and CCD semantics used here.^[5–8] Engine documentation states the implementation contract; raw runtime audit establishes which settings actually executed.

12.2 YAM and bimanual context

I2RT’s YAM and YAM Cell documentation establish the compact arm and bimanual tabletop product context; Raiden documents a YAM-oriented policy/IK stack.^[1–3] The vendored mechanical model is preserved under its adjacent license and source attribution. This task does not claim that the hand-written reference policy is the vendor controller, a learned policy, or an optimal grasp.

12.3 Why rigid contact is retained

The cube and pads are modeled as rigid. Real pad and cube deformation changes pressure distribution, torsional resistance, impact duration, and effective friction. A deformable model without pad modulus, thickness, rate dependence, cube material, mesh convergence, and force validation would add parameters without increasing justified fidelity. The rigid model is therefore retained as an explicit P0 boundary; measured compliance is a claim-promotion requirement, not a cosmetic feature to guess.

12.4 Rejected modeling claims

- **REJECTED** Parent, fixed joint, attachment, attractor, or scripted cube follow.
- **REJECTED** Kinematic robot links or direct tool-pose realization.
- **REJECTED** Tangential friction force reconstructed from a normal-only callback.

- **REJECTED** Full force closure inferred from two contact booleans.
- **REJECTED** Solver/timestep comparison with unequal post-warm-up physical state.
- **REJECTED** Calling a visual gap or hand tune “contact calibration” without measured geometry and offsets.
- **REJECTED** Treating a renderer measurement, successful episode, or simulator trace as physical data.

13 Limitations and claim-promotion triggers

Open mechanism	Current consequence	Promotion trigger
Unmeasured density/geometry	mass/inertia are priors	named cubes, uncertainty, mass/metrolgy bundle
Rigid pads/cube	pressure and torsional patch response unresolved	material/geometry data, converged deformable coupon, force validation
Unmeasured friction/restitution	slip/load history not hardware-predictive	frozen tilt, sled, drop/rebound trials across conditions
Unvalidated drive model	tracking and load may differ from hardware	joint/TCP tracking, motor/load identification, held-out paths
Normal-only callback evidence	no observed patch tangential impulse	dedicated sensor/contact API or external wrench validation
One atomic left pickup	no transport, placement, bimanual hand-off, or recovery claim	separate pre-registered tasks and falsification panels
One workstation/runtime	repeatability scope is local	second machine/runtime replication with identity manifest
Reference policy tuned by probes	not a cold-start performance claim	frozen policy benchmark versus declared baselines
No hardware safety case	simulation only	risk assessment, limits, emergency stop, supervised low-energy commissioning

Highest-priority next evidence. Populate the calibration schema with one named cube lot, pad set, table, and YAM cell; measure geometry, friction, restitution, aperture/load, and tracking; freeze parameters; then run the held-out 25-placement physical panel. Only those results can remove the “PRIOR-ONLY / NOT PHYSICALLY CALIBRATED” label.

14 Reproducibility and repository map

14.1 Primary components

Path	Responsibility
yam_cube_pick/specs.py	dimensions, rates, priors, solver fields, success thresholds, derived mass/inertia
yam_cube_pick/api.py	normalized 14-channel action and backend interface
yam_cube_pick/task.py	history-aware reward, success, failure, dwell, and integrity counters
yam_cube_pick/reference_policy.py	observable-state left-only demonstration
yam_cube_pick/isaac_backend.py	USD scene, PhysX configuration/contact monitor, cameras, audit, stepping
robots/yam/controller.py	asset adapter, physical-link read-back, exact reset, Jacobian controller, drives
yam_cube_pick/run_isaac.py	episode runner, source manifest, trajectory and frame recorder
yam_cube_pick/verify_run.py	independent saved-artifact verifier
yam_cube_pick/probes/	asset, workspace, reset, Jacobian, grasp, and numerical probes
yam_cube_pick/calibration/	JSON Schema, measurement template, and physical protocol
yam_cube_pick/evidence/	qualification, numerical, visual, video, verifier, and hash manifests

Path	Responsibility
yam_cube_pick/paper/	figures, this source/PDF, page renders, text extraction, visual QA

14.2 Core commands

Pure contracts

```
python -m pytest yam_cube_pick/tests -q
```

One canonical headless episode

```
& "$env:ISAAC_SIM/python.bat" yam_cube_pick/run_isaac.py -headless -gpu 0 -seed 3819 -output run.json
```

Complete numerical campaign

```
python -m yam_cube_pick.probes.run_numerical_campaign -isaac-python "$env:ISAAC_SIM/python.bat"
-artifact-dir yam_cube_pick/evidence/numerical_campaign/raw -log-dir logs -summary
yam_cube_pick/evidence/numerical_campaign/campaign_summary.json -force
```

Independent qualification verification

```
python -m yam_cube_pick.verify_run final_boundary_5.json final_random_20.json -expected-seeds
<frozen-list>
```

Figures and report QA

```
python yam_cube_pick/paper/make_figures.py
tectonic yam_cube_pick/paper/yam_cube_pick_physics_report.tex
python yam_cube_pick/paper/render_report.py
```

14.3 Artifact acceptance

An accepted final bundle must satisfy all of the following:

- A1. All pure tests pass from a clean Python process.
- A2. All 22 numerical artifacts are complete, share one physical source and one initial-state fingerprint, and their summary passes.
- A3. Three canonical repeats have one trajectory hash.
- A4. Physical and verifier falsifications fail for the declared reason.
- A5. Five boundary and 20 random qualification episodes pass independent verification.
- A6. Task, overview, contact, and fixed-time video evidence share the same physical source.
- A7. Figure inputs and outputs are hash-manifested.
- A8. PDF compiles without missing references/figures; every page is rendered and visually inspected; extracted text contains no pending placeholders.

15 Conclusion

The environment supports a defensible local statement: a complete, force-limited YAM articulation physically grasps and lifts a free rigid cube through finite PhysX contact, with the second arm present but non-assisting, strict history-aware success, raw contact and link-state evidence, exact controlled reset, numerical convergence/sensitivity, retained failures, fixed qualification, and realistic but nonphysical visual context. The result is stronger than an animation and narrower than a hardware prediction. Its remaining boundary is explicit: physical fidelity can advance only through the frozen calibration and held-out validation programme.

Primary source catalogue

ID	Source	Use
1	I2RT YAM documentation	robot architecture, control context, official product/model provenance
2	I2RT YAM Cell documentation	two-arm tabletop arrangement
3	Raiden documentation	reproducible YAM initialization and IK/policy context

ID	Source	Use
4	Isaac Sim 5.1 articulation controller	articulation command semantics
5	Isaac Sim simulation fundamentals	physics/render timestep separation
6	PhysX simulation documentation	TGS, iterations, friction patches, scene behavior
7	PhysX advanced collision detection	contact/rest offsets and CCD
8	Omniverse contact reports	contact points, normals, separations, normal impulses
9	Modern Robotics: force closure	grasp matrix, wrench cone, closure boundary
10	D. Stewart and J. Trinkle, “An implicit time-stepping scheme for rigid body dynamics with inelastic collisions and Coulomb friction,” 1996	nonsmooth rigid-contact time stepping
11	M. Anitescu and F. Potra, “Formulating dynamic multi-rigid-body contact problems with friction as solvable linear complementarity problems,” 1997	complementarity contact formulation
12	A. Bicchi, “On the closure properties of robotic grasping,” 1995	force/form closure theory
13	C. Ferrari and J. Canny, “Planning optimal grasps,” 1992	disturbance-based grasp quality
14	NIST SP 1227	replicable robotic-hand metrics and tests
15	NIST grasping performance	benchmarking and reproducibility
16	ISO 9283	industrial robot accuracy and repeatability framework
17	ISO/TR 13309	robot measurement guidance
18	ASTM D1894 research report	static/kinetic friction terminology and test context
19	ASTM D792	plastic density measurement reference
20	NVIDIA Omniverse PhysX source	engine provenance and inspectable implementation
21	xdofai/yam_env	vendored YAM MJCF/mesh source; adjacent local license remains authoritative

A Exact canonical configuration

Scene

- table top: 0.750 m
- table: $0.595 \times 1.300 \times 0.010$ m
- spawn: $x = [0.50, 0.70]$, $y = [-0.10, 0.10]$ m
- cube edge: 0.050 m
- cube center reset z : 0.775 m

Cube priors

- density: $1,050 \text{ kg m}^{-3}$
- mass: 0.13125 kg
- inertia: $5.46875 \times 10^{-5} \text{ kg m}^2$
- table μ_s/μ_d : 0.50/0.40
- finger μ_s/μ_d : 0.90/0.75
- restitution: 0.02

Numerics

- GPU dynamics and broadphase
- TGS; patch friction
- physics/control: 240/60 Hz
- cube iterations: 32/4
- CCD: enabled
- contact/rest: 0.30/0 mm
- sleep/stabilization: zero

Task

- lift: 0.050 m
- dwell: 0.20 s
- terminal speed: $\leq 0.15 \text{ m s}^{-1}$
- per-finger minimum normal load: 0.10 N
- timeout: 15 s
- displacement guard: 0.035 m

B Observation and action layout

Action field	Count	Meaning
left linear XYZ	3	normalized world-frame translational rate
left angular XYZ	3	normalized world-frame rotation-vector rate
left aperture	1	normalized opening-width rate
right linear XYZ	3	same interface; zero in reference

right angular XYZ	3	same interface; zero in reference
right aperture	1	same interface; zero in reference

Observation field	Count	Meaning
time	1	simulation time at control boundary
cube pose	7	position plus scalar-first quaternion
cube velocity	6	linear plus angular velocity
grippers	16	two pose/aperture records
contacts	10	contact flags, resultant/normal loads, station load
integrity	2	cube pose writes and attachments
spawn origin	3	immutable sampled cube origin

C Numerical campaign override matrix

Cell ID	Only override(s)	Expected role
baseline_repeat_0/1/2	none	independent-process repeatability
physics_120hz	physics_hz=120	coarse temporal cell
physics_480hz	physics_hz=480	refined temporal cell
position_iterations_16/64	cube_solver_position_iterations	position-solve sensitivity
velocity_iterations_2/8	cube_solver_velocity_iterations	velocity-solve sensitivity
contact_offset_150um/600um	contact_offset_m	contact-generation sensitivity
density_840/1260	cube_density_kg_m3	moderate mass/inertia sensitivity
finger_friction_lower/upper	both finger friction coefficients	moderate support sensitivity
restitution_zero/0p10	restitution	rebound sensitivity
solver_pgs	solver_type=PGS	algorithm sensitivity
ccd_disabled	enable_ccd=false	low-speed CCD sensitivity
falsification_zero_friction	all cube/finger/table $\mu_s, \mu_d = 0$	remove frictional support
falsification_heavy_cube	density 16,800 kg m ⁻³	exceed available grasp capacity
falsification_lift_200mm	required_lift_m=0.20	reject insufficient lift

D Calibration record minimums

No row may be treated as physical evidence without:

- C1.** installation, specimen, pad, table, trial, and split IDs;
- C2.** timestamp, operator, protocol revision, temperature, humidity, surface preparation, and wear state;
- C3.** instrument identity, calibration status, coordinate convention, units, uncertainty, and sample rate;
- C4.** immutable raw path and SHA-256;
- C5.** predeclared exclusion and quality flags;
- C6.** separate derived-file manifest for filtering, interpolation, synchronization, or frame transformation;
- C7.** frozen calibration, validation, and held-out membership.

E Safety and reset semantics

This is a simulator environment, not a hardware safety controller. Reset may teleport simulation state only before the episode-active flag is set. During an episode, cube pose writes are forbidden and counted. Articulation targets remain force-limited but do not constitute collision-safe hardware planning. Hardware commissioning requires independent joint/workspace limits, speed/force limits, collision monitoring, emergency stop, risk assessment, and supervised low-energy trials.

F Evidence bundle checklist

Artifact	Required content
final_boundary_5.json	five complete qualification records, runtime audits, full trajectories
final_random_20.json	twenty complete frozen random records
campaign_summary.json	22 cells, one source, one initial state, exact repeats, deltas, failures
alignment_gate_sweep_seed588.json	four matched phase-gate cells with complete initial state, decision snapshots, forces, and outcomes
final_visual_seed3819.json	task-camera trace and capture records
final_overview_seed3819.json	overview-camera sequence
final_contact_seed3819.json	contact-camera sequence
final_video_seed3819.mp4/json	fixed-time encoded motion plus decode/hash audit
qualification_summary.json	independent 25-record acceptance report
final*_verification.json	independent task, overview, and contact capture acceptance reports
artifact_manifest.json	paths, roles, sizes, and SHA-256 values
figure_manifest.json	exact JSON inputs and generated figure hashes
yam_cube_pick_physics_report.pdf	compiled dossier
report_contact_sheet.jpg	visual QA of every PDF page
report_qa.json	PDF, text, contact-sheet, and per-page hashes plus pending-placeholder gate